

2 SFIA Review

This section reviews my professional skills development under the Skills Framework for the Information Age (SFIA).

Data Science

CYCLE NUMBER		1	
START DATE		10/08/2024	
EXPECTED DURATION		8 Months	
SKILL ACRONYM	SKILL LEVEL	DESCRIPTION	NARRATIVE
DATS	Level 4	Investigates the described problem and dataset to assess the usefulness of data science and analytics solutions. Applies a range of data science techniques and uses specialised programming languages. Understands and applies rules and guidelines specific to the industry, and anticipates risks and other implications of modelling. Selects, acquires and integrates data for analysis. Develops data hypotheses and methods and evaluates analytics models. Advises on the effectiveness of specific techniques	See section 2.2

		based on project findings and comprehensive research. Contributes to the development, evaluation, monitoring and deployment of data science solutions.	
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After the first stage of my placement, I collaborated closely with a colleague who had initially modelled willow cuttings geometrically, treating the plants as cone-shaped objects to predict how many 200 millimetre long and 8 to 22 millimetre wide cuttings each willow could yield based on its overall dimensions. He introduced this geometric approach, struggled to accommodate the irregularity of the willow form, and produced inconsistent results. Recognising that machine learning (ML) could uncover hidden patterns and non-linear relationships to reduce such structural modelling limitations, I proposed using our existing measurement dataset to experiment with ML techniques.

When I joined the project, we had only data from a single harvest, and my initial machine-learning models yielded low accuracy, with Coefficient of Determination (R^2) values below 0.5. Coefficient of Determination (R^2) indicates the proportion of variance in the dependent variable explained by the model, with values closer to 1 representing a better model fit:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \text{ where } \bar{y} \text{ is the mean of observed values. Due to the}$$

dataset's limited size, I conducted new experiments after presenting these early findings at team meetings. Expand the new features in my data frame, such as biomass weight from past harvests.

By January 2025, we had collected measurements across three harvests. I noticed that our data was still separated across multiple Excel sheets with inconsistent formats, and that further cleaning was necessary. Collaborating with the colleague who is working on geometric modelling, I designed and implemented a comprehensive data-cleansing pipeline in Python. I then conducted exploratory data analysis and visualisations, which I presented progress on at weekly team meetings. These cleaned and combined data were used, which included key features such as biomass weight and counts of

cuttings from the first and second-largest stems on a plant. I applied various machine-learning models. These efforts proved fruitful: our Random Forest model achieved an R^2 of 0.86 and a mean absolute error (MAE) of less than 1, indicating a significantly improved fit and practical utility for deployment.

This workflow, which involves selecting and integrating disparate datasets, defining analytical hypotheses, applying specialised data-science techniques, evaluating and refining models, and developing a deployment-ready ML solution, directly aligns with the Level 4 descriptor. It demonstrates how I investigated problems and anticipated risks in modelling and contributed to monitoring and deploying a viable data-science solution. This reflects the core responsibilities of a Data Science practitioner in an industry setting.

Programming/software development

CYCLE NUMBER		1	
START DATE		15/09/2024	
EXPECTED DURATION		10 Months	
SKILL ACRONYM	SKILL LEVEL	DESCRIPTION	NARRATIVE
PROG	Level 4	Designs, codes, verifies, tests, documents, amends and refactors complex programs/scripts and integration software services. Contributes to the selection of the software development methods, tools and techniques. Applies agreed standards and tools to achieve well-engineered outcomes.	See section 2.3

		Participates in reviews of own work and leads reviews of colleagues' work.	
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The first programme I developed after joining the project was the automated specific leaf area measurement tool described earlier. 'Specific leaf area (SLA; leaf area/ leaf mass) is an important parameter to plant growth modellers because it determines how much new leaf area to deploy for each unit of biomass produced.' SLA measurements provide valuable data for our project. Measuring each image manually using ImageJ took five to ten minutes per image. To address this, I learned to use OpenCV (cv2) within Python, recreating the manual steps. First, convert pixel-to-millimetre scaling, automatic thresholding and leaf area selection in each image. Then, to streamline the process. Initially, I encountered several issues, including non-uniform pixel selection, noise in thresholded images, and contours incorrectly detecting all white areas as leaves (as shown in Figure 2.1). These challenges echoed common errors in binary image segmentation, where thresholding can produce false contours without proper tuning. To solve this, I leveraged the fact that all 500+ images shared a consistent setup: they are all taken at identical camera distances and feature a yellow measurement tape by the side. I redesigned the scaling step to require the user to input the length of this yellow tape, allowing accurate pixel-to-millimetre calculations. I then constrained contour detection to green areas likely to be leaf objects, significantly improving accuracy and reducing false detection. Based on my software development experience, I also revised the SLA image-capture protocol during the final harvest. I reorganised the leaf arrangements to ensure apparent gaps and neater alignment, allowing better separation of individual leaves in each photograph. As a result, I collected the final set of SLA image data, and the new capture method significantly enhanced compatibility with the tool developed, achieving cleaner binary masks and more accurate leaf-area measurements.

With non-technical users in mind, the next step is prioritising ease of use. Packaging this solution into a Graphical User Interface built with OpenCV dependencies often fail. So, after trying to develop an executable, I opted to host the code in a private GitHub (cloud-based code storage) repository and integrated it with Streamlit. This ensured internal access only, avoiding additional costs while maintaining security and usability.

Binary Threshold



Figure 2.1: Binary Threshold Image of leaves

Following the development of the leaf-area tool, I created the next software to address a time-consuming limitation in downloading irrigation data from our systems, which we monitor via BlueLab devices. BlueLab is a manufacturing company that offers solutions to measure pH, Conductivity (EC), and Temperature, which we installed in our experimental setup. And that data is stored on their Edenic online platform. Edenic's web interface requires manually selecting precise start and end timestamps with minute accuracy for each device and download session. With over ten systems' data to monitor quarterly, this manual process was prone to errors and inefficiencies, risking data gaps and inconsistencies. After speaking with our on-site staff, I investigated Edenic's capabilities and discovered that a REST API can be provided with API keys and telemetry endpoints. I gained access to our organisation's Edenic account and created an API key to fetch data. During testing, I attempted to retrieve three months of data at once but encountered missing data. To address this, I studied the documentation provided by Edenic and found that the telemetry endpoint limits each request to 500 data points. After discussing with the team and realising our analysis only required trend-level resolution, I limited the interval and fit three months of data in 500 rows. The team confirmed this approach was acceptable afterwards.

To ensure accessibility for non-technical colleagues, I packaged the script using Gooney (Python library that helps in creating graphical user interfaces (GUIs) for command-line applications), creating a simple GUI application named BlueLab

API Data Downloader (as shown in Figure 2.2), where the users are able to enter a start and end date, and the tool automatically downloads via the Edenic API, saving each system's data in Comma-Separated Values (CSV) files titled with system number and date range. This executable runs locally, requires no coding knowledge, and enforces structured data output that preserves completeness and consistency across devices.

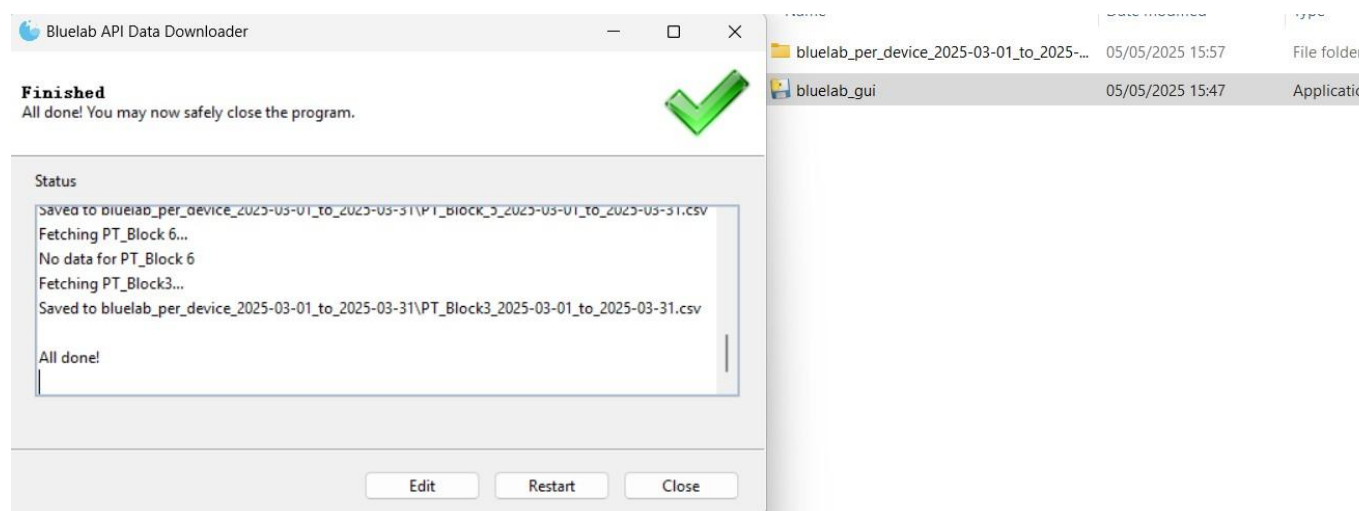


Figure 2.2: BlueLab API Data Downloader

The Taeda Tech project is designed to be a three-year project, and my placement fell in the project's final year. As the project approached its final year, staffing on-site was reduced. As the on-site nutrient monitoring and weekend checks remained necessary, I realised that much of this monitoring involved interpreting readings from the BlueLab device. I recognised an opportunity to augment site operations through AI-enhanced analysis. Although Edenic API does not provide real-time updates, I realised we could retrieve and process the last data reading through a large language model to offer actionable advice for site operations. I designed and implemented an automated advisory tool built on the Edenic API and Google's Gemini-1.5 model, named Taeda Assistant.

Follow the design architecture shown in Figure 2.3. I began by writing Python functions to list devices and fetch the most recent telemetry summaries collected from BlueLab devices. Next, I integrated Google's generative AI Software Development Kit (SDK), which supports calling Gemini-1.5 for text-based summarisation. A fine-tuned system prompted the model to behave like a plant science assistant. It dynamically calls and returns summarised data, turning it into actionable insights, for example, advising when the nutrient solution pH is below the target. (As shown in Figure 2.4).

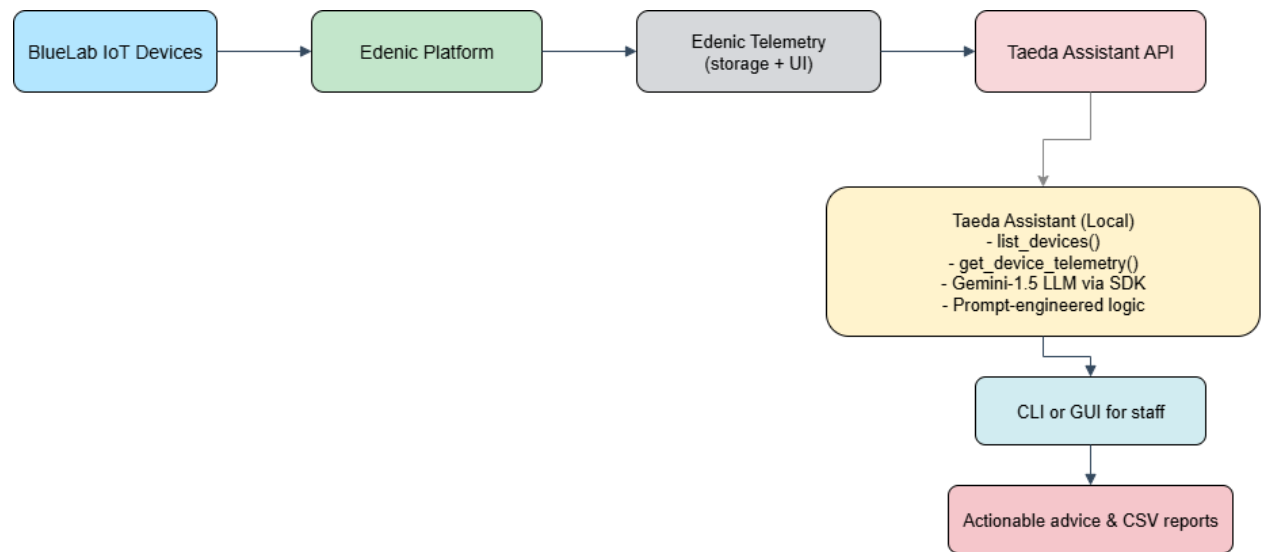


Figure 2.3: Taeda Assistant architecture diagram

You: That's ok, if best PH is 7, ec is 1.5. Any suggestion or action we should take?

 **Taeda Assistant**

Based on the current data, the pH of 4.20 is significantly lower than the optimal range of 7. The electrical conductivity (EC) of 1.20 is also slightly lower than the ideal 1.5. Both of these readings suggest the nutrient solution might be too acidic and possibly lacks sufficient nutrients.

****Suggested Actions:****

1. ****Adjust pH:**** Gradually raise the pH of the nutrient solution using a pH up solution. Monitor the pH closely and make small adjustments to avoid shocking the plants. Regular monitoring and adjustments will be needed to maintain the optimal pH.
2. ****Increase EC:**** Add more nutrients to increase the EC towards the target of 1.5. Be cautious not to over-fertilize. Follow the fertilizer instructions carefully and consider using a diluted solution at first.
3. ****Monitor:**** Continue monitoring the pH and EC levels regularly (at least daily) and adjust as needed. Observe the plants closely for any signs of stress or nutrient deficiencies (e.g., yellowing leaves, stunted growth). Adjust your nutrient regimen based on the plants' response.
4. ****Consider the cause:**** Investigate potential causes for the low pH and EC, such as issues with the nutrient solution mixing, reservoir contamination, or problems with the irrigation system. Clean and maintain the system regularly to prevent further issues.

****Important Note:**** I am an AI and cannot provide specific instructions on how to adjust pH or EC, or what nutrients to use. Always refer to your plant species' specific nutrient requirements and follow the instructions of your chosen fertilizers and pH adjusting products carefully. Improper adjustments can harm the plants.

Figure 2.4: Example of Prompt from the Taeda Assistant

If the project continues to grow, this software can be expanded and fine-tuned to suit larger deployments. The Taeda Assistant provides an internal, cost-effective solution for the team, providing operational oversight, reducing manual workload, and supporting future budgeting decisions.